



OPEN Unveiling interactions of spatial-temporal information in tactile motion perception

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Tactile perception is inherently dynamic, relying on active manual exploration to extract information about motion and surface properties. Spatiotemporal inputs facilitate tactile motion perception by conveying information both direction and speed perception. Although previous studies have examined these features separately, the interactions between spatial and temporal features in shaping perceptual outcomes remain poorly understood. To address this gap, we conducted two psychophysical experiments in which tactile motion stimuli, varying in direction, speed and spatial frequency (wavelength), were delivered to the distal fingerpad of healthy participants, and then requested the participants to report their feedback directly. In Experiment I, we found that the anisotropic distortion of directional perceptual bias is quadrant-dependent, while variations in speed did not alter this general pattern. Experiment II revealed a dissociation between spatial and temporal contributions to perception. Spatial frequency primarily determined the overall pattern of perceptual bias, reflecting the structural properties of the stimulus. In contrast, speed modulates its dynamic expression by influencing the amplitude and phase of deviations. Additional psychometric function analyses indicated that tactile speed perception arises from a combination of linear and nonlinear processes. Collectively, these findings elucidate how the brain integrates spatiotemporal cues to construct a coherent tactile motion representation, thereby accounting for the systematic directional distortions and nonlinear speed estimation.

Keywords Touch, Tactile motion perception, Somatosensory, Psychophysics

Tactile motion perception is fundamental to recognizing textures^{1–3} and manipulating objects^{4–6}, which is crucial for everyday tasks and navigation^{7–9}. Despite significant advances in understanding tactile perception, the mechanisms underlying the integration of spatial and temporal features in tactile motion perception remain poorly understood.

Research has shown that cortical neurons exhibit firing rate-based tuning to global motion direction^{10,11}, while peripheral afferents track spatiotemporal patterns for local perception^{12–15}. This mismatch in encoding can result in perceptual bias¹⁶, where the perceived direction and speed of motion may not align with the actual physical stimulus. A previous study addressed this issue by dividing perceptual bias into systematic and nonsystematic biases. Systematic bias refers to global deviations in perceived direction attributable to somatotopic and allocentric frames of reference, while nonsystematic bias encompasses the dynamics of stimulation, resulting in a model emphasizing the relative weight of proprioceptive influences on sensory input¹⁷.

The way we perceive tactile speed is shaped by the interplay between how fast a stimulus moves and the texture of its surface. Textured patterns provide vibration and pressure changes that influence the frequency and intensity of the perceived speed¹⁸. Neurophysiological studies have highlighted a complex interplay between both speed-sensitive and non-sensitive neurons¹⁹, challenging our understanding of the neural encoding of tactile motion^{20,21}. Despite speed being a crucial perceptual feature, significant gaps remain in understanding

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the neural mechanisms underlying tactile motion perception, the interactions between the spatial and temporal features, and the role of motion in active touch^{22,23}.

In the present study, we investigated how spatial and temporal features of tactile stimuli jointly influence tactile motion perception. To this end, we conducted two psychophysical experiments in which tactile motion stimuli of varying direction, speed, and spatial frequency were delivered to the distal fingerpad, and participants reported their percepts through a touchpad interface. Experiment I assessed the role of stimulus speed on directional perception, whereas Experiment II examined how speed and spatial frequency interact to shape tactile motion perception. Our results revealed anisotropic distortions in directional judgments and nonlinear estimation in speed perception, highlighting a dissociation between spatial and temporal contributions during spatiotemporal integration. These findings provide new insight into the neural mechanisms that govern how spatiotemporal interactions construct tactile motion perception.

Results

To explore perceptual bias and the integration of spatial-temporal information in tactile motion perception, we designed an experimental setup to present a diverse set of tactile motion stimuli, encompassing various spatial and temporal features. These stimuli were presented on the left index fingerpad, and participants reported the motion trajectory using their right index finger on a touchpad (Fig. 1a). The lab-designed tactile stimulator enabled us to covary direction, speed, and spatial frequency, allowing for the exploration of potential interactions of spatial and temporal features in tactile motion perception (Fig. 1b). We recruited 33 participants and allocated them into two separate groups for distinct experiments. In Experiment I (Fig. 1c), 17 participants were tested to explore anisotropies in perceptual bias for different directions of motion on the distal fingerpad. In Experiment II (Fig. 1d), 16 participants were examined to study the effects of spatial and temporal features on tactile motion speed perception. Additionally, mathematical approaches were employed to gain insights into the potential roles and interactions of these features in tactile motion perception.

Anisotropic distortion in the perception of tactile motion direction

In Experiment I, designed to investigate perceptual bias for different directions of motion and the influence of speed on tactile motion perception, the left index finger was stimulated with 24 directions under two speed conditions, covering the entire fingerpad (Fig. 1c). Traces drawn by the participants were pooled together following a feature extraction procedure (see Data Processing in Methods). Directional perceptual bias (B_d),

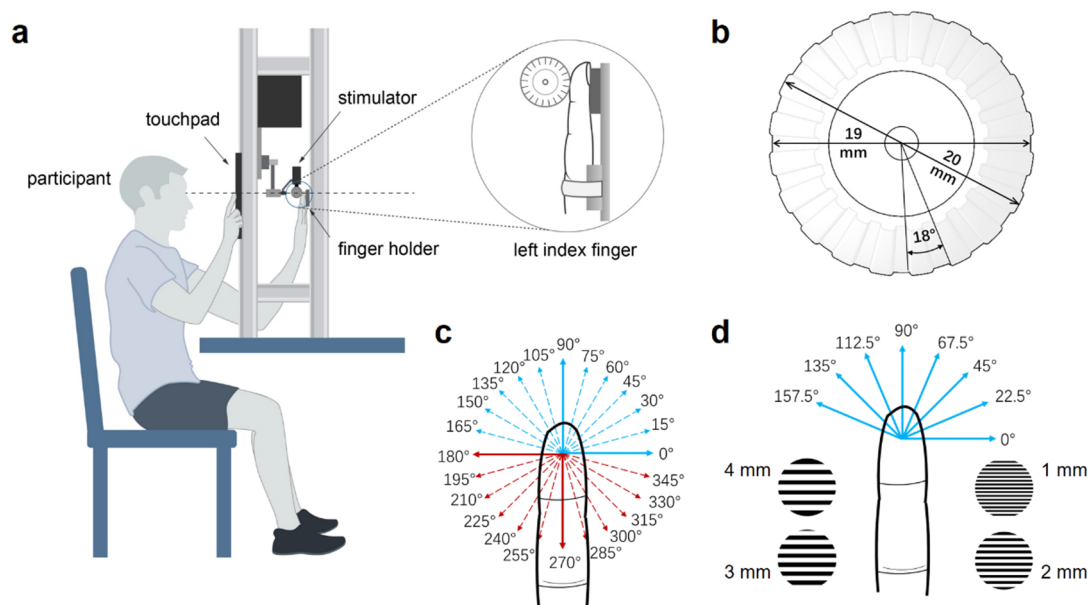


Fig. 1. Setup and experiments design. **(a)** The participant is seated with their left index finger securely positioned, while using their right hand to indicate the perceived direction and speed of the tactile motion stimulus on the touchpad. The left index finger is held vertically within the apparatus, and the initial position of stimulus ball (sphere) is set on the tangent plane between the fingerpad and the ball for each participant. **(b)** A schematic representation of a stimulus ball with a 3-mm wavelength is illustrated. The ball has a diameter of 20 mm and the grating indentation is 0.5 mm. The wavelength is defined as the maximum interval between the gratings, which are separated by an 18° step for a 3-mm wavelength ball. The angle of gratings is 5.62, 11.25, and 22.5° for 1-mm, 2-mm, and 4-mm wavelength balls, respectively. **(c)** Tactile motion directional symmetry experiment utilized a comprehensive range of stimulus directions, spanning from 0° to 360° with 15° intervals. **(d)** In the tactile motion speed integration experiment, eight motion directions were included, ranging from 0° to 157.5° with 22.5° intervals. Additionally, six distinct speeds (20, 40, 80, 160, 240, 320 mm/s) were used for each of the four wavelengths of the stimulus ball (1, 2, 3 and 4 mm).

representing the deviation between the perceptual and veridical direction of a stimulus, was calculated, and further analysis was conducted to evaluate the effect of speed and potential distal fingerpad anisotropies.

In Fig. 2a, perceptual biases (B_d) are depicted for the stimulus speeds of 40 mm/s (blue line) and 240 mm/s (red line) plotted against stimulus direction. Perceptual bias values consistently ranged from 0° to -45° , indicating a clockwise rotational shift. The bias tended to decrease as stimulus directions shifted from a horizontal clockwise to a vertical direction, exhibiting a known oblique effect^{24,25}. Interestingly, increasing stimulus speed did not significantly alter the overall anisotropy. A two-way ANOVA (2 speeds $\times$ 24 directions, $n = 17$) revealed no significant main effect of speed on anisotropy ($F(1, 16) = 0.034$, $p = 0.857$, $\eta^2 = 0.002$) and no significant interaction effect between speed and direction ($F(23, 368) = 1.384$, $p = 0.114$, $\eta^2 = 0.322$). However, there was a significant main effect of direction on perceptual bias ($F(23, 368) = 7.593$, $p < 0.001$, $\eta^2 = 0.080$), indicating that the bias varies systematically across directions. Despite this, we defined the mean perceptual bias, averaged across directions, as systematic bias B_s . This systematic bias remained consistent across speeds, with no significant difference between the 40 mm/s and 240 mm/s conditions: $-16.64^\circ \pm 0.82^\circ$ SEM for 40 mm/s, $-16.46^\circ \pm 1.02^\circ$ for 240 mm/s (paired t-test $t(16) = 0.184$, $p = 0.857$, Cohen's $d = 0.045$), indicating its consistency to changes in stimulus speed.

However, perceptual biases exhibited more pronounced differences across fingerpad quadrants, as shown in Fig. 2b. There was an overall reduced perceptual bias in the second quadrant (Q2, upper left) and fourth quadrant (Q4, lower right), whereas there was an overall greater perceptual bias in the first quadrant (Q1, upper right) and third quadrant (Q3, lower left). A one-way ANOVA revealed significant modulation in the anisotropy pattern (Fig. 2b, $F(3, 48) = 9.004$, $p < 0.0001$, $\eta^2 = 0.360$), and post-hoc tests showed most pairwise differences between quadrants were significant (Bonferroni: Q1-Q2, $p < 0.0001$; Q1-Q4, $p < 0.0001$; Q2-Q3, $p = 0.012$; Q3-Q4, $p = 0.002$). In contrast, comparisons between Q1-Q3 ($p = 0.6077$, $BF_{10} = 1.67$) and Q2-Q4 ($p = 0.8128$, $BF_{10} = 0.94$) were non-significant, with Bayesian factors indicating only anecdotal evidence for differences between these quadrant pairs. These results suggest a central symmetry in the distribution of perceptual bias. Supporting this, perceptual bias across opposite directions was significantly correlated (Fig. 2c, Pearson correlation, $r = 0.679$, $p = 0.02$). Supplementary experiments on the right index finger (Fig. S1) and the left index finger in a horizontal setup (Fig. S2) confirmed similar patterns of anisotropic distortion and systematic biases, demonstrating the robustness of the observed effects regardless of dominant hand or apparatus orientation.

The role of speed and spatial frequency on tactile motion direction perception

In Experiment II we used stimuli with varying spatial frequencies (ball wavelength), rotating at different speeds to investigate the influence of spatial and temporal features on tactile motion speed perception. To streamline conditions, we stimulated only 8 directions of motion covering the distal aspects of the fingerpad (Fig. 1d) and four ball wavelength conditions, each presented at six speeds. In the analysis we included variables for systematic bias and nonsystematic bias, and mathematics models were employed to illuminate the effects of spatial frequency and velocity on speed perception.

Systematic biases at stimulus speeds of 20 mm/s, 40 mm/s, 80 mm/s, 160 mm/s, 240 mm/s, and 320 mm/s were -13.76° , -14.68° , -13.08° , -11.96° , -11.34° and -11.44° , respectively (represented by the different colored solid lines in Fig. 3A that run parallel to the dashed zero bias line). Overall, systematic bias showed no significant difference between the stimulus speeds ($F(5,75) = 1.676$, $p = 0.151$, $\eta^2 = 0.100$) and averaged $-12.71^\circ \pm 0.61^\circ$, indicating a clockwise shift as observed in Experiment I.

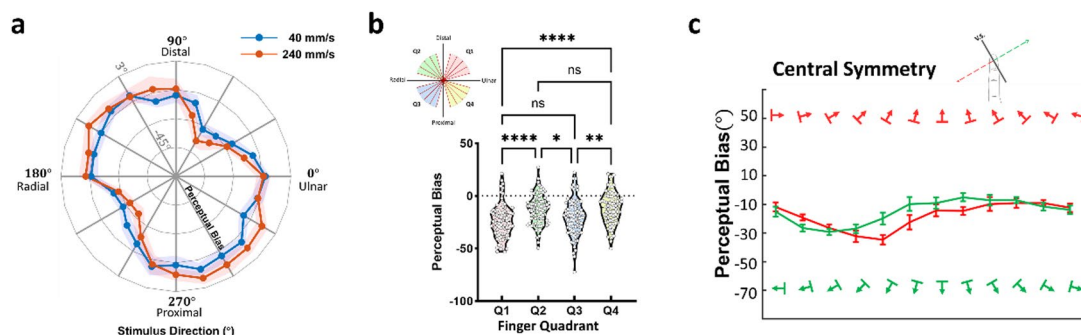


Fig. 2. Anisotropic distortion in the perception of tactile motion direction. **(a)** Dots indicate the average direction bias (B_d) of participants ($n = 17$) across varying motion directions at each speed (40 mm/s in blue, 240 mm/s in red). Radial axes represent the magnitude of perceptual bias, with concentric gray circles indicating bias values. The scale spans from -45° (inner circle) to 3° (outer circle) to encompass the full range of the observed bias data. Anatomical references (ulnar, distal, radial, proximal) indicate the directional orientation on the stimulated hand. Shaded regions around the data points represent the standard error of the mean (SEM). **(b)** Stimulus directions were grouped into four quadrants. The violin plots illustrate perceptual bias distributions for varying directions as recorded from each participant. The location on the finger for each quadrant is shown in the subplot. Pairwise analysis indicates that perceptual biases for diagonal quadrants were similar, contrasting with those of adjacent quadrants (ANOVA $F(3,48) = 9.004$, $p < 0.0001$, $\eta^2 = 0.360$; post-hoc Bonferroni, * represents $p < 0.05$, ** represents $p < 0.01$, **** represents $p < 0.0001$). **(c)** The plots depict the average directional bias for central symmetries (red and green) among subjects.

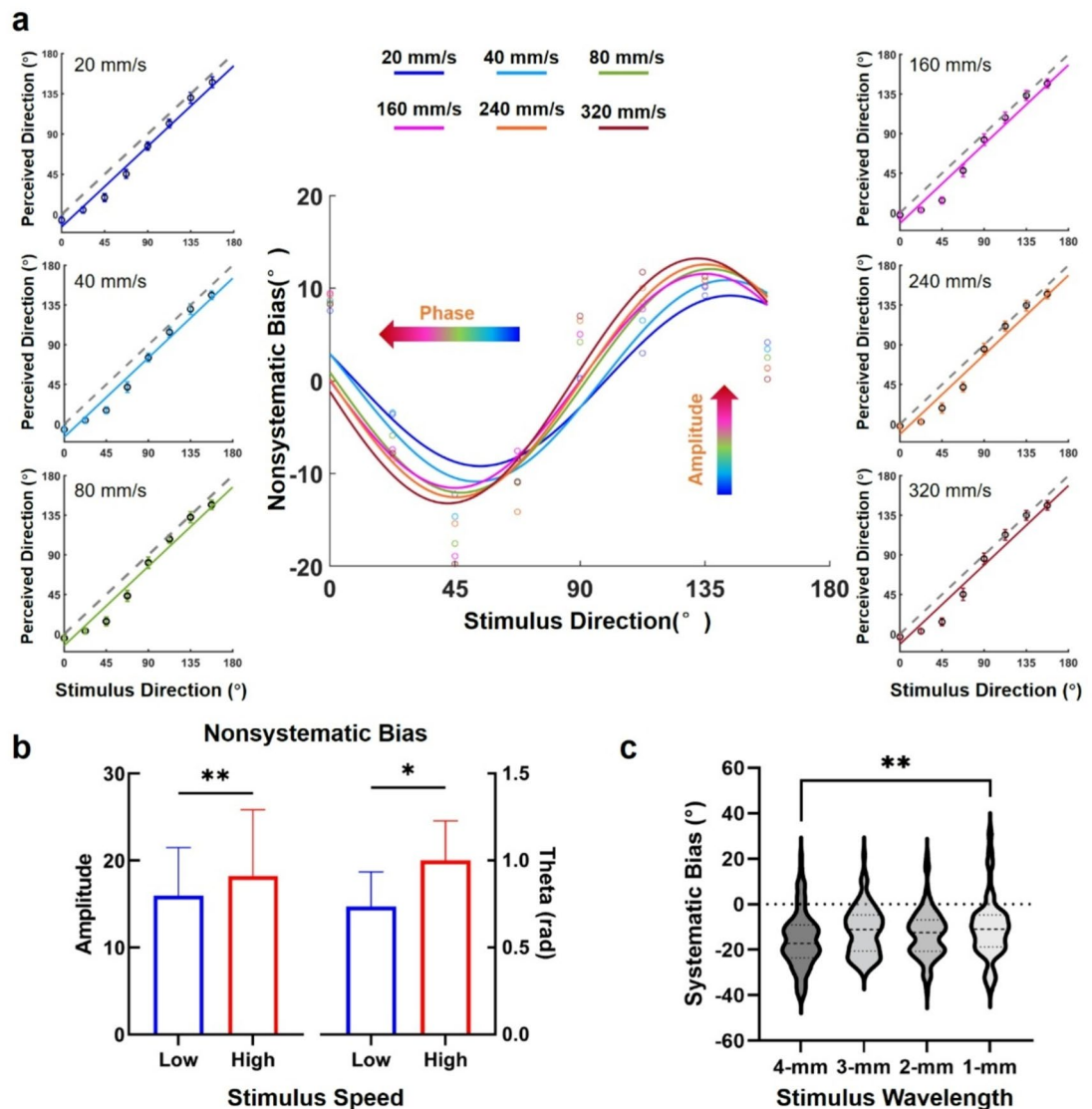


Fig. 3. The role of speed and spatial frequency on tactile motion direction perception. **(a)** Perceived direction (black circles) of tactile motion for the variety stimulus speeds (20, 40, 80, 160, 240, 320 mm/s) presented in two columns of graphs. Different colored solid lines represent linear fits to systematic bias for different stimulus speeds, while the black dashed line represents the axis of symmetry between perceived direction and stimulus direction. In the middle graph, nonsystematic bias is illustrated with color dots, each associated with different cosine fitting curves. The data is further divided into low-speed (20, 40 and 80 mm/s stimulus speeds) and high-speed (160, 240 and 320 mm/s) subgroups. **(b)** The amplitude of nonsystematic bias shows a significant difference between low-speed and high-speed subgroups (t -test, $t(15) = 3.014$, $p < 0.01$), also for phase of the nonsystematic bias (t -test, $t(15) = 2.702$, $p < 0.05$). Data is presented as the mean $\pm$ the standard error of mean (SEM). **(c)** Systematic bias varied moderately across stimulus wavelengths. Linear mixed-effects modeling revealed a significant effect of wavelength ($p = 0.006$), with a notable difference between the 1-mm and 4-mm conditions (Bonferroni-corrected, $p = 0.011$). Dashed lines represent the median and quartile lines (25th and 75th percentiles) in each violin plot.

Nonsystematic bias was fitted using a cosine function (as defined in Eq. 4) for various stimulus speeds. The peak amplitudes of the nonsystematic bias (A) were 15.49, 15.85, 16.58, 18.42, 18.00, and 18.21°, while the corresponding phases (θ_p) were 0.366, 0.748, 1.09, 0.79, 0.963, and 1.251° for stimulus speeds ranging from 20 mm/s to 320 mm/s. Both the amplitude and phase of the nonsystematic bias exhibited a slight upward trend with increasing speed. Statistical analysis revealed a significant main effect of speed on phase (ANOVA, $F(5,75) = 7.518$, $p < 0.0001$, $\eta^2 = 0.334$), whereas no significant main effect of speed was found on amplitude (ANOVA, $F(5,75) = 2.291$, $p = 0.054$, $\eta^2 = 0.133$). These results highlight the dynamic modulation of tactile motion perception by temporal stimulus properties.

To explore potential influence of stimulus speeds on tactile motion direction perception, data were categorized into low-speed and high-speed groups based on a validated binary effect (details in Supplementary, Fig. S3).

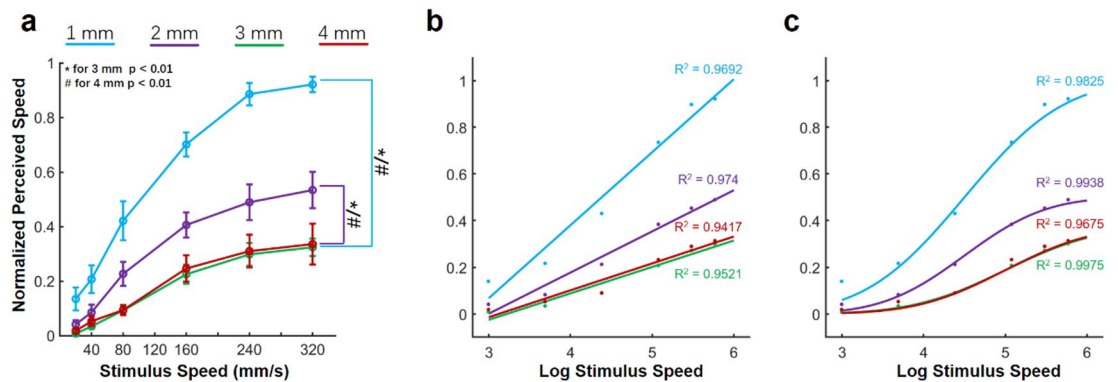


Fig. 4. Effect of spatial features and velocity on tactile motion speed perception. **(a)** The plot shows the normalized perceived speeds as a function of stimulus speed. It demonstrates an increase in perceived speed with increasing stimulus speed that increases with ball wavelength. Compared with the 1-mm and 2-mm ball wavelengths, the perceived speeds evoked by the 3-mm and 4-mm ball wavelengths were significantly lower. Data is presented as mean $\pm$ standard error of mean (SEM). Two fitting curves were conducted to analyze these data: **(b)** linear function, and **(c)** Gaussian distribution-based psychometric function.

The low-speed group consisted of data from stimulus speeds of 20 mm/s, 40 mm/s, and 80 mm/s, while the high-speed group comprised data from 160 mm/s, 240 mm/s, and 320 mm/s. The amplitude of nonsystematic bias showed significant difference between the two groups (High-speed Group = 18.21 ± 0.76 , Low-Speed Group = 15.98 ± 0.64 ; $t(15) = 3.014$, $p = 0.009$, Cohen's $d = 0.754$), while the phase of nonsystematic bias only exhibited a trend (High-speed Group = 0.74 ± 0.10 , Low-speed Group = 1.00 ± 0.10 ; $t(15) = 2.702$, $p = 0.016$, Cohen's $d = 0.675$), as shown in Fig. 3b. In addition, supporting the findings of Experiment I, no significant difference was observed in the systematic bias between speed groups (High-speed Group = -11.58 ± 0.88 , Low-speed Group = -13.84 ± 0.83 , $t(15) = 1.813$, $p = 0.090$, Cohen's $d = 0.453$; see Supplementary Fig. S4a).

Beyond stimulus speed, we further examined the impact of the stimulus ball's wavelength. As shown in Fig. 3c, the wavelength of the stimulus ball had a moderate effect on systematic bias, with the overall ANOVA showed no significance ($F(3, 45) = 2.358$, $p = 0.084$, $\eta^2 = 0.136$). However, a linear mixed-effects model accounting for individual differences revealed a significant effect of wavelength on systematic bias ($p = 0.006$). In line with this, a descriptive post-hoc comparison revealed a notable difference between the 1-mm and 4-mm wavelength conditions (Bonferroni-corrected, $p = 0.011$). In contrast, ball wavelength did not significantly affect nonsystematic bias, including amplitude ($F(3, 45) = 0.156$, $p = 0.925$, $\eta^2 = 0.010$) nor did it show a trend for phase ($F(3, 45) = 1.345$, $p = 0.272$, $\eta^2 = 0.082$), as shown in Supplementary Fig. S4b.

Taking together, these findings suggest that temporal features, represented by scanning speed, play a role in influencing nonsystematic bias, while spatial features, represented by the wavelength of the stimulus ball, impact systematic bias. This indicates that significant and complementary roles of both temporal and spatial features in shaping tactile motion direction perception.

The effect of speed and spatial features on tactile motion speed perception

We further investigated the impact of spatial features (ball wavelength) on tactile motion speed perception. Perceived speed was significantly modulated by the stimulus ball wavelength ($F(3,45) = 29.236$, $p < 0.001$, $\eta^2 = 0.661$), as shown in Fig. 4a. Specifically, 1-mm and 2-mm wavelength balls elicited higher normalized perceived speeds compared to 3-mm and 4-mm wavelength balls (Bonferroni, $p < 0.001$ for 1-mm ball vs. 3-mm or 4-mm ball, $p < 0.05$ for 2-mm ball vs. 3-mm or 4-mm ball). The effects of wavelength reached a lower plateau with 3-mm and 4-mm wavelength balls showing almost identical curves. This suggests the existence of a spatial resolution threshold. For speeds above 20 mm/sec and spatial frequencies between 1- and 3-mm, spatial features play a significant role in modulating tactile motion speed perception where a 3-mm wavelength ball potentially represents the upper limit of spatial features capable of impacting speed perception.

We used both linear and nonlinear functions to model the relationship between perceived speed and stimulus velocity and wavelength. The linear model was used to assess whether there was a simple, monotonic increase in tactile motion speed perception, as showed in Fig. 4b. Despite achieving high R-square values exceeding 0.94, the unequal distribution of the sum of squares errors (SSE) suggested the linear fit did not fully capture the effect of spatial features on perceived speed (see Fig. 4b and Supplementary Table S1 and Table S2). To gain a more comprehensive understanding of the role of spatial features in tactile motion speed perception, a Gaussian distribution-based psychometric function²⁶ was used as a nonlinear descriptive model (Eq. 6, Supplementary Materials), with the results of the fit depicted in Fig. 4c. This model provided an averaged R-square of 0.98, indicating a superior fit compared to the linear models (details in Supplementary, Table S1). The Gaussian fit revealed three aspects of the psychometric function for tactile motion speed perception: an approximately linear increase at lower speeds, heightened sensitivity to speed changes at intermediate speeds (around 90 mm/s), and saturation at high speeds. These results suggest that the Gaussian distribution-based psychometric function offers a more comprehensive computational framework for understanding tactile motion speed perception, involving nonlinear mechanisms that extend beyond linear trends.

Discussion

The present study aimed to unravel the intricate interplay of spatial-temporal information in tactile motion perception. By investigating directional symmetry in tactile motion processing, exploring interactions between spatial and temporal features, and proposing a quantitative model for tactile-motion speed perception, we sought to enhance our understanding of tactile motion processing. One of the key findings of our study is that the anisotropic distortion of perceptual bias for the distal fingerpad is quadrant-dependent, rather than specific to a preferred direction of motion. Furthermore, we discovered that spatial and temporal features have divergent influences on tactile motion perception. Temporal features exert their influence on nonsystematic bias, while spatial features might be associated with systematic bias. Moreover, our results revealed that the processing of tactile motion speed encompasses both linear and nonlinear components, each associated with specific speed ranges. Collectively, these findings underscore the distinct roles played by spatial and temporal features in the integration of information crucial for tactile motion perception.

A novel observation in our study was polar angle asymmetry in tactile perception of the direction of motion of the fingertip. Anisotropy in tactile perception, indicating a broad oblique effect, is well-established regardless of body position^{16,27,28} and environmental settings^{29,30}. Our study expands on the existing knowledge of tactile directional perception by examining directional motion across a full 360 degrees. We observed significant increases in perceptual biases in the first quadrant (Q1, upper right) and third quadrant (Q3, lower left), contrasted with more accurate responses in the second quadrant (Q2, upper left) and fourth quadrant (Q4, lower right). These observations align with Kuroki's findings^{28,31}, in which stimuli inclined inwards led to more perceptual distortion (corresponding Q1 and Q3 in our study). However, Kuroki found this distortion only on the arm, and neither location nor the size of receptive field could explain its absence on the palm. In our study, we propose that the distortion results from the integration of tactile and proprioception inputs, influenced by natural hand placement posture and tactile exploration. Finger and arm posture^{32,33}—such as the natural 45-degree angle of the arm relative to the body during rest—tends to align the hand more closely with the horizontal plane in body-centered (egocentric) coordinates. This natural positioning effectively rotates the orientation of the fingers, aligning the Q2–Q4 axis closer to the horizontal axis. Furthermore, when exploring the texture of an object with the hand, movement along the Q2–Q4 axis on the fingertip is preferred over the Q1–Q3 axis. Understanding this directional sensibility sheds light on the interaction between mechanoreceptor distribution, neural processing, and hand posture in shaping perceptual biases.

Tactile polar angle asymmetries differs from visual polar angle asymmetry^{34–37}, where visual anisotropy is generally better along the horizontal axis and the lower vertical axis³⁸. Visual polar angle asymmetries are influenced by factors such as photoreceptor density, retinal ganglion cells^{39,40}, and cortical tissue distribution^{41–44} in both humans and monkeys. In contrast, research on perceptual asymmetry in tactile studies is limited, often assumed to be related to the distribution of skin mechanoreceptors³⁰ and their corresponding neurons in somatosensory cortex. These polar angle asymmetries suggested inherent limitations in both tactile and visual system regarding location-specific sampling, restricting the amount of information transmitted from peripheral receptors to the brain. Furthermore, we observed that polar angle asymmetries may exhibit a unique form of symmetry in the Cartesian coordinate system. Our findings reveal that tactile perceptual performance exhibited central symmetry during direction discrimination. In contrast, visual perception exhibits Y-axis symmetry⁴⁵. The difference between tactile and visual perceptions may be attributed to variations in neural circuits and computational processing models.

To comprehend spatiotemporal interactions, we categorized perceptual bias into systematic and nonsystematic biases as proposed by Chen et al.¹⁷. Systematic bias serves as a reliable indicator of the relative posture between different body parts, reflecting multisensory integration. Interestingly, we observed a notable difference in systematic bias between coarse (4-mm ball) and dense (1-mm ball) spatial wavelengths of the tactile stimulus (Fig. 3c), with the greatest bias occurring at the lowest spatial frequency. It was consistent with Kuroki and Nishida findings⁴⁶, where they emphasized that tactile motion perception results from both low-level vibrations and high-level feature tracking. This suggests that spatial features, rather than temporal features, influence systematic bias.

In contrast, nonsystematic bias likely originates from neural directional preferences or receptor responses, retaining dynamic information. Our findings reveal specific effects on the amplitude and phase of nonsystematic bias during tactile motion perception. Notably, stimulus speed significantly affected both amplitude and phase, with clear variation on amplitude and significant shift on phase (Fig. 3b). These findings suggest that temporal features can distort direction perception, consistent with previous research which indicated that faster arm movements induced larger localization errors⁴⁷. Accounting for nonsystematic bias revealed distinct roles for spatiotemporal information in tactile perception. These distinct mechanisms alter directional perception and spatial accuracy indicating that spatiotemporal integration relies on the dynamic interplay of spatial configuration and temporal features.

Tactile speed perception is a complex process dependent on the interplay of spatial and temporal stimulus features, underpinned by neural mechanisms^{48–50}. Our study demonstrated that perceived speed depends on the spatial feature (ball wavelength) and stimulus speed (Fig. 4a). Higher spatial frequency wavelengths increase the range of perceived speeds, suggesting precise textures provide more detailed information about surface texture and facilitate easier feature tracking. However, there is a spatial limit for this influence, as 3-mm and 4-mm ball wavelengths lead to similar speed perception, potentially related to the two-point discrimination thresholds and mechanoreceptor densities on the fingerpad⁵¹. These observations are consistent with prior research on tactile speed discrimination results^{19,50}, which showed that coarse textured (low spatial frequency) surfaces are perceived as slower. This suggests that perceived speed may not always directly reflect actual speed but is instead modulated by surface properties. For example, smooth-surfaced stimuli result in increased speed discrimination errors⁵², highlighting the role of surface in modulating tactile speed perception. Furthermore, tactile speed

perception is not solely dictated by peripheral mechanoreceptor distribution, as different afferents, particularly PC afferents, exhibit preferences for specific spatial frequency ranges¹⁹. Similar effects are reported in the visual domain, where spatial frequency influences speed perception^{53,54}. These findings suggest that motion perception across sensory modalities may share common computational principles⁵⁵, with spatial frequency playing a critical role in shaping perceptual interpretations.

Consistent with previous research^{20,21,50}, our results confirm a monotonic increase in perceived speed with stimulus speed, applicable to speeds below 160 mm/s and extending to higher speeds (Fig. 4a). At higher speeds, perceived speed reaches a plateau regardless of ball wavelength. Although a linear function between stimulus and perceived speeds fits well, significant deviations of the psychophysical measures at higher speeds indicates that linear function failed to capture the data's complexity. This suggests that tactile speed perception involves nonlinear neural coding. Nonlinear processes are common across different sensory modalities. Visual speed perception, for example, involves both linear and nonlinear computations across different brain areas (e.g., retina, primary visual cortex, and middle temporal areas), enabling accurate speed perception^{56–58}. Similarly, speed perception in the tactile modality is shaped by the interaction of various mechanoreceptors, which transmit distinct types of spatial and temporal information⁵⁹. Recent findings by Merz et al.⁶⁰ reveal that both visual and tactile modalities exhibit nonlinear effects of speed on the perception of onset and offset positions, explained by the speed prior account. This prior-driven mechanisms likely increases complexity of motion perception, especially in the tactile domain, where sparse innervation density⁶¹ and long afferent delays⁶² impose additional challenges. These factors result in slower and less precise transmission of tactile information, adding to the overall intricacies of speed perception in the tactile modality.

Given the spatial and temporal interactions in tactile speed perception, we introduced a Gaussian distribution-based psychometric function to incorporate both spatial and temporal features into a computational model (Table S1 and Fig. 4c). This function proposes a nonlinear integration of spatial and temporal information, resulting in improved goodness-of-fit values of the psychophysics. This approach differs from a previous study by Greenspon et al.⁴⁹, which identified nonlinear relations in the power spectral densities of vibrations and speed, suggesting an interaction between spatial and temporal features in speed perception. Our findings indicate that the proposed function performs better with high spatial features (with 1-mm and 2-mm wavelength balls) compared to low spatial features (3-mm and 4-mm wavelength balls). The reduced range of perceived speeds is attributed to the lower elicited vibration by larger wavelengths⁵⁰. This suggests that there may be limitations in discriminating the speed of coarse surfaces within a specific range.

The diversity of textures and shapes in the real-world results in high-dimension tactile sensations, and several studies have explored the role of motion in dynamic touch for tactile perception. Although our present study has provided valuable insights into the intricate relationship between spatial and temporal features in tactile motion perception, it is important to acknowledge certain limitations in our experimental design. For example, we used a simplified parameter “ball wavelength” to represent how stimulus texture interacts with tactile motion processing. Furthermore, our study used a rotating tactile stimulus on a stationary finger to examine the integration of spatial and temporal features. Although these methods may not fully capture the complexities of real-world tactile experiences, our findings are intriguing and merit further investigations into their neural underpinnings. Future research endeavors that incorporate more natural textures and modes of tactile interaction may shed additional light on the intricate mechanisms of tactile motion perception.

In conclusion, our study advances the understanding of tactile motion processing by revealing quadrant-dependent anisotropic distortion, central symmetry, and the distinct roles of spatial and temporal features in shaping perceptual biases. The proposed Gaussian distribution-based psychometric function provides a nuanced framework for comprehending the nonlinear integration of spatial and temporal information in tactile speed perception. Further investigations into these mechanisms, considering the complexities of real-world tactile interactions, will contribute to a more comprehensive understanding of tactile motion perception.

Methods

Participants

This study was conducted with the approval of the Ethics Committee of the Second Affiliated Hospital of Zhejiang University School of Medicine (Approval No. 2015-081), and all methods were performed in accordance with the relevant guidelines and regulations. All participants gave their written informed consent prior to taking part and were treated in accordance with the Declaration of Helsinki (2008, version 6).

Thirty-three individuals were recruited and randomly divided into two groups. One for a tactile motion direction symmetry experiment ($n=17$, mean age: 22.13 ± 2.75 years, seven females) and the other for a tactile motion speed integration experiment ($n=16$, mean age: 26.18 ± 1.94 years, seven females). All participants were right-handed, as determined by the Edinburgh Handedness Inventory.

Stimuli apparatus and procedure

Participants were seated with their left forearm comfortably resting on holders, as shown in Fig. 1a. The left forearm was vertically aligned with the elbow set at 90° angle, and the wrist, hand, palm up, and index finger (D2) were secured in a consistent posture to facilitate tactile stimulation. Participants' head, eyes, and index finger were aligned along the posterior-to-anterior axis. A touchpad (PTH-660, Wacom Com Ltd., Japan) was placed between the participants and the tactile stimulator. Participants used their right index finger to draw an oriented line on the touchpad to indicate the perceived direction and speed of tactile stimuli. To mask any noise generated by the tactile stimulator, participants wore headphones emitting white noise.

Tactile stimuli were administered using a lab-designed ball stimulator, as described in the study by YC Pei et al.⁶³, which has demonstrated its effectiveness and reliability. This stimulator provided three degrees of freedom, facilitated by three independent motors (Faulhaber Inc., Switzerland), enabling the delivery of tactile

motion stimuli in various directions and speeds. Specifically, the stimulator uses a stepper motor (AM1524) for precise depth control, while two servo motors (1024M006S and 1516E006SR) enable rotation of the ball and its directional adjustment. Each stimulus ball had a diameter of 20 mm (Fig. 1b) and featured four distinct wavelengths of square-wave gratings, each with the depth set to 500 μm and a duty cycle of 45%. To ensure a continuous perception of tactile motion, the speed and direction of a stimulus were initiated before contact. The ball was then lowered at a speed of 5 mm/sec to achieve a 0.5-mm indentation, and stimulation lasted for 1 s before the ball was lifted off from the finger. Before an experiment, the stimulus ball contacted the fingerpad to determine the starting position. The direction of motion was delineated within a planar coordinate system, with the upward direction (proximal to distal) along the index finger defined as 90° , as shown in Fig. 1c.

Participants were instructed to respond by drawing oriented line on a touchpad to indicate direction and speed, reported by the length of the line. A point at the center of the touchpad marked the initial position of the reporting trajectory, and a circular region with 5-cm radius around the center served as a boundary. Following each stimulus trial, an auditory cue (a beep) prompted participants to report their tactile motion perception. The trajectory of the drawn oriented lines was recorded using the Psychophysics Toolbox Version 3 (PTB-3, <http://psychtoolbox.org>) in MATLAB (R2018b, MathWorks, Natick, MA) at a sampling rate of 200 Hz. The perceived speed of the stimuli was calculated based on the number of pixels involved in the trajectories.

Experimental design

This study was designed to investigate the perceptual bias for different directions of motion, and the influence of spatiotemporal features on tactile speed perception. To this end, two experiments were conducted as below.

Experiment I: tactile motion directional symmetry

The primary objective of this experiment was to examine perceptual bias symmetry of the distal fingerpad in the perception of the tactile motion direction and assess the impact of speed on this bias. A stimulus ball with a 4-mm wavelength was used to present stimuli in 24 directions, ranging from 0° to 360° in 15° increments, at stimulus speeds of 40 mm/s and 240 mm/s, as shown in Fig. 1d. Each specific combination of stimulus parameters was presented five times, resulting in a total of 240 trials (24 directions $\times$ 2 speeds $\times$ 5 repetitions). To ensure reliability, the trials were divided into five blocks, each corresponding to a repetition of the stimuli. Participants had the flexibility to take breaks between these blocks, and the entire session took approximately 40 min. Stimuli were presented in a pseudo-random order.

Experiment II: tactile motion speed integration

Expanding on the insights gained from *Experiment I*, *Experiment II* introduced covarying spatial-temporal features to investigate their influence on tactile motion speed perception. Four stimulus balls with four different wavelengths (1, 2, 3, and 4 mm), six distinct speeds (20, 40, 80, 160, 240 and 320 mm/s), and eight directions ranging from 0 to 157.5° with 22.5° increments were employed, as shown in Fig. 1d. This comprehensive experiment included 960 trials (8 directions $\times$ 6 speeds $\times$ 5 repetitions $\times$ 4 wavelengths) and took about 180 min to complete. Participants had the flexibility to complete these sets over two days, with a 3-minute rest interval after every 48 trials (8 directions $\times$ 6 speeds) to prevent attention fatigue.

Data processing

During each trial, the positional time series of the oriented line drawn on the touchpad was recorded. This recorded signal was differentiated to calculate the perceived speed, while direction was calculated using the arctangent function, $\tan^{-1}(Y/X)$, for the entire trajectory. In addition, an average perceived direction and speed was computed based on the five stimulus repetitions.

Perceptual bias B_d was defined as the difference between the perceived direction D_p and the stimulus direction D_s :

$$B_d = D_p - D_s \quad (1)$$

For the left hand, a negative B_d would imply a clockwise shift in direction perception, indicating that participants perceived the direction as being rotated clockwise compared to its actual stimulus direction. Conversely, a positive B_d points to a counterclockwise shift in direction perception.

Perceptual bias was separated into systematic bias and nonsystematic bias, where systematic bias refers to global deviations in perceived direction attributable to somatotopic and allocentric frames of reference and nonsystematic bias encompasses the dynamics of stimulation. The systematic bias B_s was defined as the mean of the perceptual bias B_d :

$$B_s = \frac{1}{n} \sum_{i=1}^n B_d \quad (2)$$

where n was the number of stimulus directions. The nonsystematic bias, B_{ns} , calculated the difference of the perceptual bias for each motion direction from the overall perceptual bias¹⁷. The calculation can be expressed as follows:

$$B_{ns} = B_d - B_s \quad (3)$$

In our study, a cosine function was employed as the fitting curve:

$$B_{ns} = A \cos(mD_s + m\theta_p) \quad (4)$$

where m was set to 2, as in a previous study¹⁷, to ensure a best fitting curve for our dataset. Parameters A and θ_p represent the peak amplitude and peak phase of the nonsystematic bias, respectively.

To analyze tactile motion speed perception, perceived speed was calculated as the ratio of the trajectory length of the drawn line to the recording duration. The trajectory length was manually verified to exclude periods of initial hesitation or instances where the pointer exceeded screen boundary issues ensuring that only stable and deliberate drawing phases were included. Perceived speed was standardized using Z-score normalization across participants and conditions to facilitate statistical comparisons. And both linear and nonlinear models were used to explore interactions between stimuli and perceived speed. For linear fitting, the equation adopted was as follows:

$$S_e = k \times \log(S_p) + c \quad (5)$$

where S_e was the estimated normalized perceived speed, S_p , corresponded to the stimulus speed, and k denoted the slope and c denoted the intercept of the linear model.

To incorporate presumed mechanisms underlying the tactile motion perception process, a Gaussian distribution-based psychometric function²⁶ was used to fit the data:

$$S_e = \frac{1}{2^w} \times \left[\operatorname{erf}\left(\frac{\log(S_p - \mu)}{\sqrt{2\delta^2}}\right) + 1 \right], \quad (w = 1, 2, 3, 4) \quad (6)$$

where w was the wavelength of the stimulus ball, μ was the center threshold of perceived speed, and δ was free parameter corresponding to the just-noticeable difference, and erf was Gaussian error function. This model postulates that stimulus wavelength is a predominant factor influencing speed perception, and μ was set as 90 mm/s based on the value associated with the highest R-square (details in Supplementary, Fig. S3). Both linear and nonlinear models were implemented using *Curve Fitting Toolbox in MATLAB* (2021a) for regression analysis.

Statistical analysis

All the statistical analyses were conducted using SPSS software (version 22, IBM Corp., Armonk, NY). Normality was assessed using the Shapiro-Wilk test and Levene test. In Experiment I, a two-way repeated-measures ANOVA tested whether exists interaction effects of stimulus speed and direction on the perceptual bias anisotropy. Then we further explore the main effect of speed and direction respectively. Paired t-tests assessed speed effects on the mean perceptual bias across, while one-way repeated-measures ANOVA investigated anisotropic patterns within each quadrant. Pearson's correlation evaluated symmetry in perceptual bias. In Experiment II, repeated-measures ANOVA was used to assess the impact stimulus speed on direction perception, including systematic bias, amplitude of nonsystematic bias, and phase of nonsystematic bias. ANOVAs were also performed to assess the effects of stimulus wavelengths on tactile direction perception. To explore the effects of wavelength on tactile speed perception, ANOVA was first used to examine the difference as the stimulus speed varied, followed by a Pearson's correlation to assess the consistency between the curves.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Conceptualization, B.Q. and HYL.; Methodology, B.Q. and X.T.; Investigation, B.Q., X.T., Z.T., H.W.; Writing – Original Draft, B.Q., K.E.S. and HYL.; Writing – Review & Editing, B.Q., R.M.F. and HYL.; Visualization, B.Q. and L.L.; Supervision, HYL.; Funding Acquisition, HYL. and G.P.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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